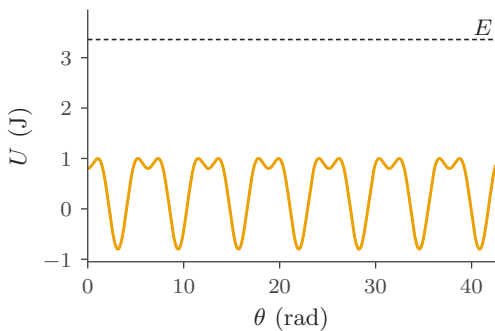
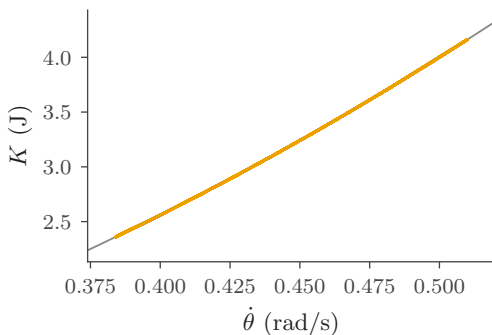


Case 4:  $\theta_0 = 0$ ,  $\dot{\theta}_0 = 0.4$  rad/s;  $k = 0.1$  N/m,  $L = 4$  m,  $M = 2$  kg,  $g = 0.1$  m/s<sup>2</sup> ( $\mu = +1.00$ )

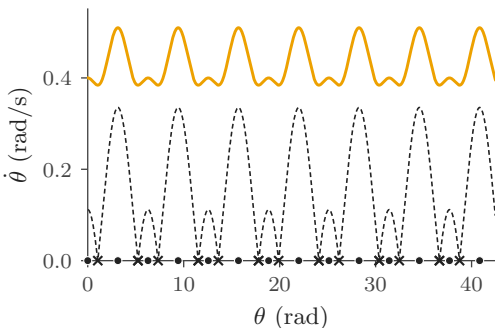
Potential energy



Kinetic energy



Phase diagram



Position

